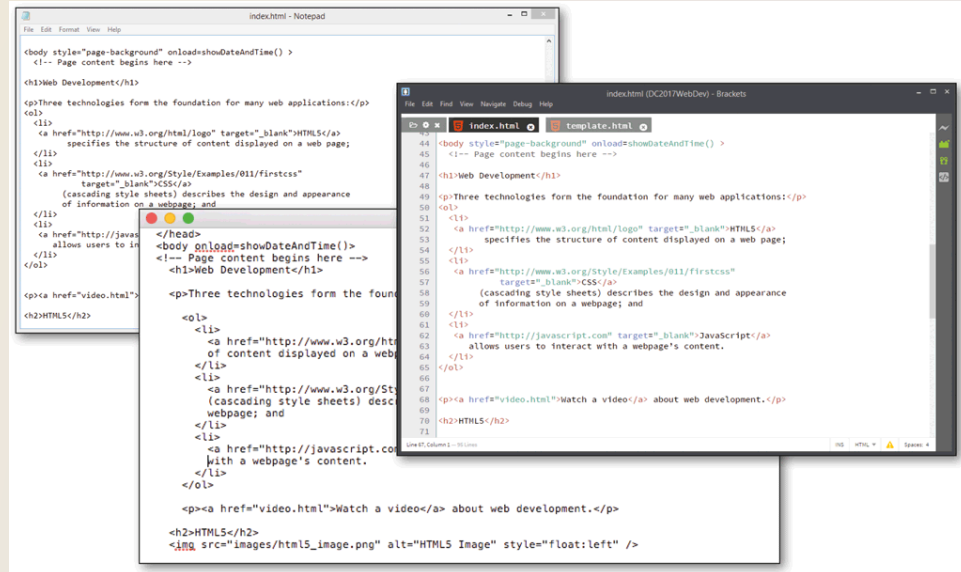


Discovering Computers Enhanced Edition ©2017

Tools, Apps, Devices, and the Impact of Technology

Focus On Web Development



Objectives Overview

Discuss tools for creating a website, such as text editors, code editors, and content management systems, and when to use each

Explain the uses of HTML5, CSS, and JavaScript technologies when creating websites

Discuss concepts related to web development, including static and dynamic content, relative and absolute references, HTML tags and attributes, and embedded and inline styles

Explain how to view a webpage's source code after displaying the page in a browser

Objectives Overview

Explain the unique role of the index.html page in a website

Use HTML tags to add a title, headings, paragraphs, images, links, ordered and unordered lists, and videos to a webpage

Use CSS to specify fonts, colors, and styles for text and background images or colors for webpages

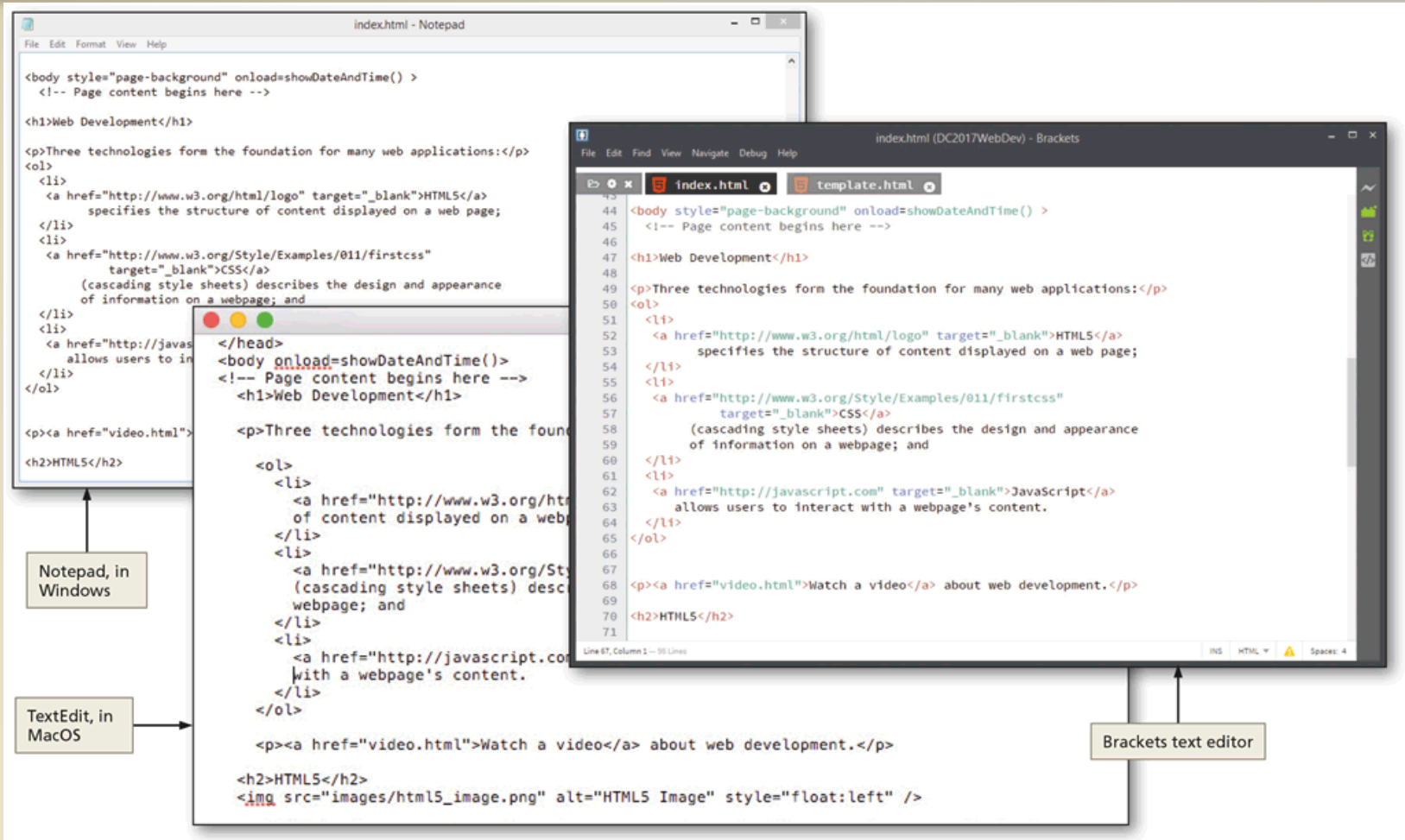
Use JavaScript to display the current date and time on a webpage

Upload a website to a web server using an FTP program

Tools for Creating a Website

- Text editors
 - Similar to a word processing program, but lacks most text formatting features
 - Operating systems typically include a text editor
 - A code editor is a type of text editor that contains additional features to help web developers write code

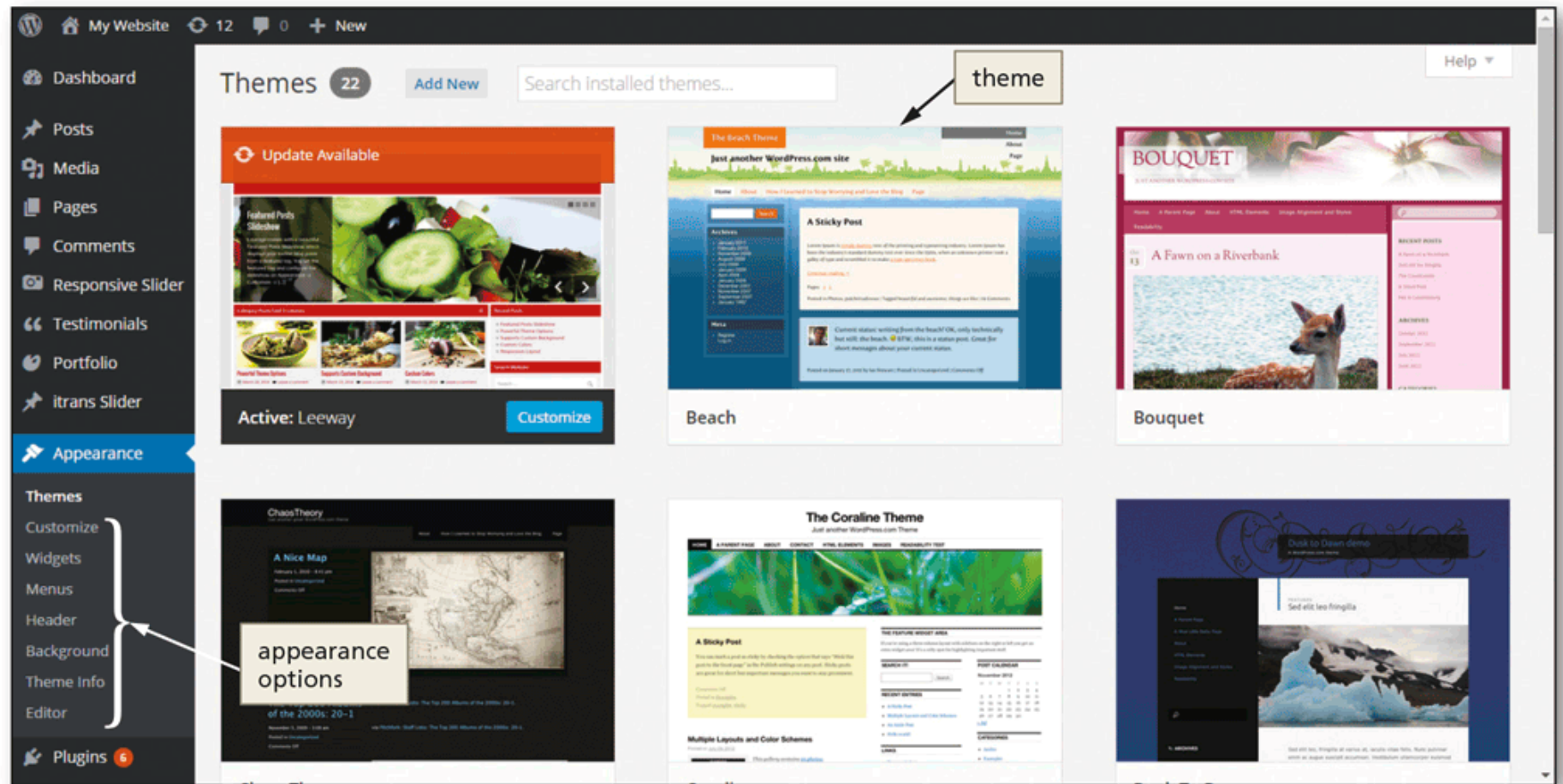
Tools for Creating a Website



Tools for Creating a Website

- Content Management Systems
 - Enables and manages the publishing, modification, organization, and access of various forms of documents and other files on a network or the web
 - Web developer creates a theme, and one or more website content administrators enters the content

Tools for Creating a Website



Website Technologies

- Hypertext Markup Language (HTML) uses a set of codes called tags to format documents for display in a browser
- A complementary technology called cascading style sheets (CSS) contains specifications for the fonts, colors, layout, and placement of these HTML elements on a webpage
- JavaScript is a programming language for creating programs that a browser can run to generate content for a website

Website Technologies

The image displays a screenshot of the Smithsonian National Zoological Park website, specifically the 'Meet Our Animals' page for 'Great Apes & Other Primates'. The website features a navigation bar with links like 'Visit', 'Animals', 'Membership', 'Science', 'Education', and 'Support'. A sidebar on the left lists 'GREAT APES & OTHER PRIMATES' with sub-links: 'Meet the Primates', 'Facts', 'Exhibit', 'News', 'Science', and 'Enrichment'. The main content area has a large heading 'GREAT APES & OTHER PRIMATES' followed by a sub-heading 'GORILLAS' and a paragraph about western lowland gorillas. Below this is a section 'PRIMATES AT THE ZOO' with a paragraph about the zoo's primate collection and an image of a gorilla. The bottom section is 'ABOUT PRIMATES' with a paragraph about primate species and another paragraph about the pygmy mouse lemur. A call to action at the bottom left says 'Together, We Save Species \$49.50 million'.

Annotations with arrows point to various elements on the website and their corresponding HTML code:

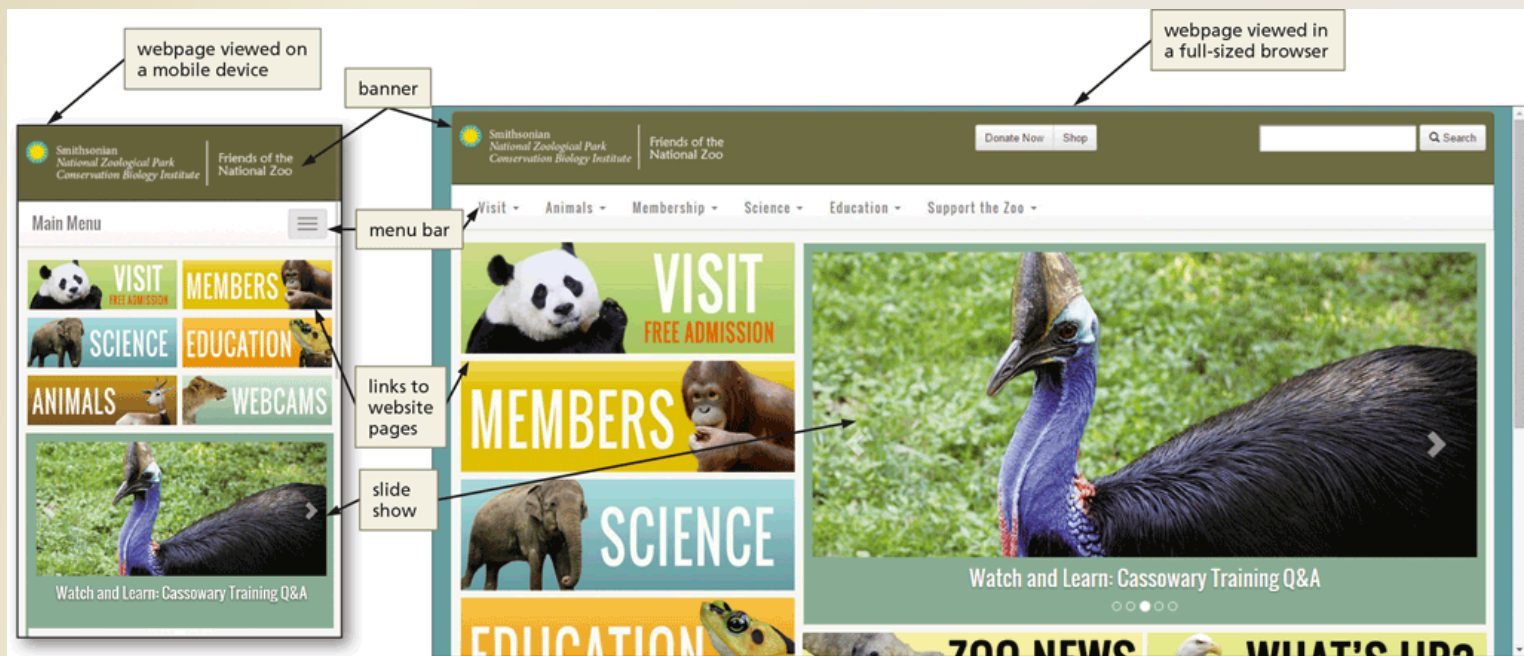
- link to external style sheet**: Points to the `<link href="https://fonts.googleapis.com/css?family=Oswald:400,300,700" rel="stylesheet">` tag in the HTML code.
- h1 header**: Points to the `<h1>Great Apes & Other Primates</h1>` tag in the HTML code.
- h2 header**: Points to the `<h2>Gorillas</h2>` tag in the HTML code.
- hyperlink**: Points to the `<a href="/Animals/Primates/MeetPrimates/MeetGorillas/">` tag in the HTML code.
- h3 header**: Points to the `<h3>Primates at the Zoo</h3>` tag in the HTML code.
- image**: Points to the `` tag in the HTML code.
- paragraph**: Points to the paragraph of text about the zoo's primate collection in the HTML code.

The HTML code shown on the right is as follows:

```
1 <!DOCTYPE html
2 PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
3 "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
4 <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en"><!-- Inst
5 codeOutsideHTMLIsLocked="false" -->
6 <head>
7 <title>Great Apes and Other Primates - National Zoo</title>
8 <link href="https://fonts.googleapis.com/css?family=Oswald:400,300,700" rel=
9 />
10 </head>
11 <body>
12 <h1>Great Apes & Other Primates</h1>
13 <h2>Gorillas</h2>
14 <p>
15 <a href="/Animals/Primates/MeetPrimates/MeetGorillas/">Six western lowland
16 youngest is Kibibi, born in 2009.
17 </a>
18 </p>
19 <h3>Primates at the Zoo</h3>
20 <a href="http://www.flickr.com/photos/nationalzoo/3707340844/" title="Goril
21 
23 <p>
24 The Zoo is home to many primates. Orangutans and western lowland gorillas
25 tamarins, Geoffroy's marmosets, and howler monkeys, can be found in the
26 Gibbon Ridge and lemurs at <a href="/animals/primates/meetprimates/meetlemur
27 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />
28 </p>
29 <p>
30 On mild days, the orangutans can sometimes be seen overhead as they travel
31 Ape House and Think Tank. </p>
32 <h3>About Primates</h3>
33 <p>There are 376 species of primates in the world&#8212;from humans and apes
34 </p>
35 <p>
36 The smallest primate is the pygmy mouse lemur, which can fit in the palm of
37 pounds. Most primates live in warm climates, and most depend on forests for
38 src="/admin/images/icons/img_arrowdk.gif" alt="" width="15" height="11" />
39 </p>
40 </body>
41 </html>
```

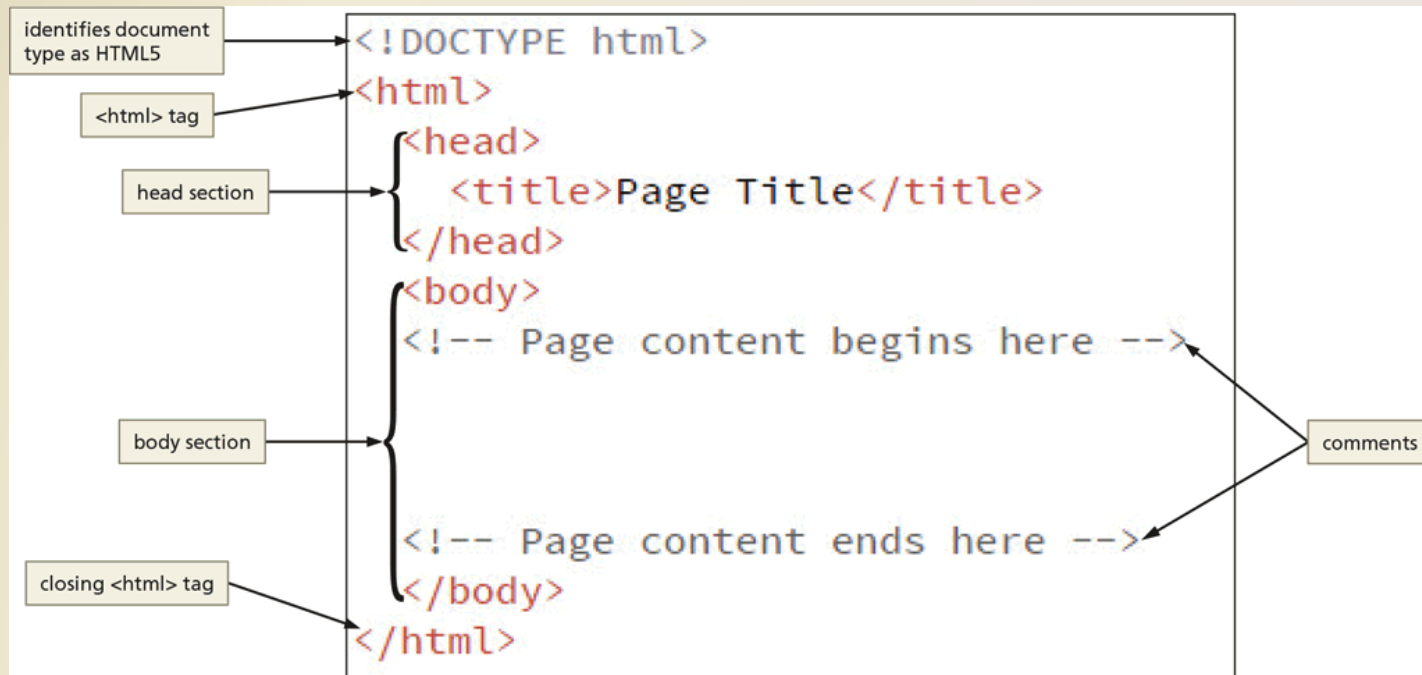
Website Technologies

- Responsive webpages automatically adjust the size of their content to display appropriately relative to the size of the screen of the device on which it is displayed



Structure of a Webpage

- A webpage's source code contains text marked up with HTML tags that instruct a browser how to display that content



Structure of a Webpage

- The World Wide Web Consortium (W3C) oversees the specification of HTML Standards
- The W3C provides a free, online HTML5 validator application to ensure that a webpage's HTML tags follow the specifications, or rules, for HTML5

Structure of a Webpage

The screenshot displays the Nu Html Checker interface. The title bar reads "Nu Html Checker". Below it, a note states: "This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change". The main heading is "Showing results for contents of text-input area".

The "Checker Input" section contains a "Show" dropdown with options for "source" (selected), "outline", and "image report", along with an "Options..." button. The "Check by" dropdown is set to "text input". The input area contains the following HTML code:

```
<h1>Web Development</h1>
Three technologies form the foundation for many web applications:</p>
<ol>
  <li>
    <a href="http://www.w3.org/html/logo" target="_blank">HTML5</a>
    specifies the structure of content displayed on a web page;
  </li>
  <li>
    <a href="http://www.w3.org/Style/Examples/011/firstcss"
    target="_blank">CSS</a>
    (cascading style sheets) describes the design and appearance
    of information on a webpage; and
  </li>
</ol>
```

A callout box labeled "<p> tag is missing" points to the closing tag of the paragraph in the input code.

Below the input area is a "Check" button. The "Message filtering" section shows three messages:

1. Info: The Content-Type was text/html. Using the HTML parser.
2. Info: Using the schema for HTML with SVG 1.1, MathML 3.0, RDFa 1.1, and ITS 2.0 support.
3. Error: No `p` element in scope but a `p` end tag seen.
From line 50, column 66, to line 50, column 69
Locations: `</p>`

A callout box labeled "validator warning message" points to the third message in the list.

Creating the index.html File

- Run the text editor of your choice
- Navigate to and open the template.html file
- If necessary, enable the word wrap feature so that you can view all the webpage text without scrolling horizontally
- Save the file using the file name, index.html. Do not exit the text editor

Creating the index.html File

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Page Title</title>
5  </head>
6  <body>
7    <!-- Page content begins here -->
8
9
10
11    <!-- Page content ends here -->
12  </body>
13 </html>
```

webpage structure

Copying the Starter Text from a Source File and Pasting It in the index.html File

- Open the startertext.txt file
- Select all the text in the startertext.txt file
- Copy the text
- Close the startertext.txt file
- Display the text editor window containing the index.html file
- Position the insertion point on blank line after the line that says, `<!--Page content begins here -->`
- Paste the text
- Save the changes to the index.html file

Copying the Starter Text from a Source File and Pasting It in the index.html File

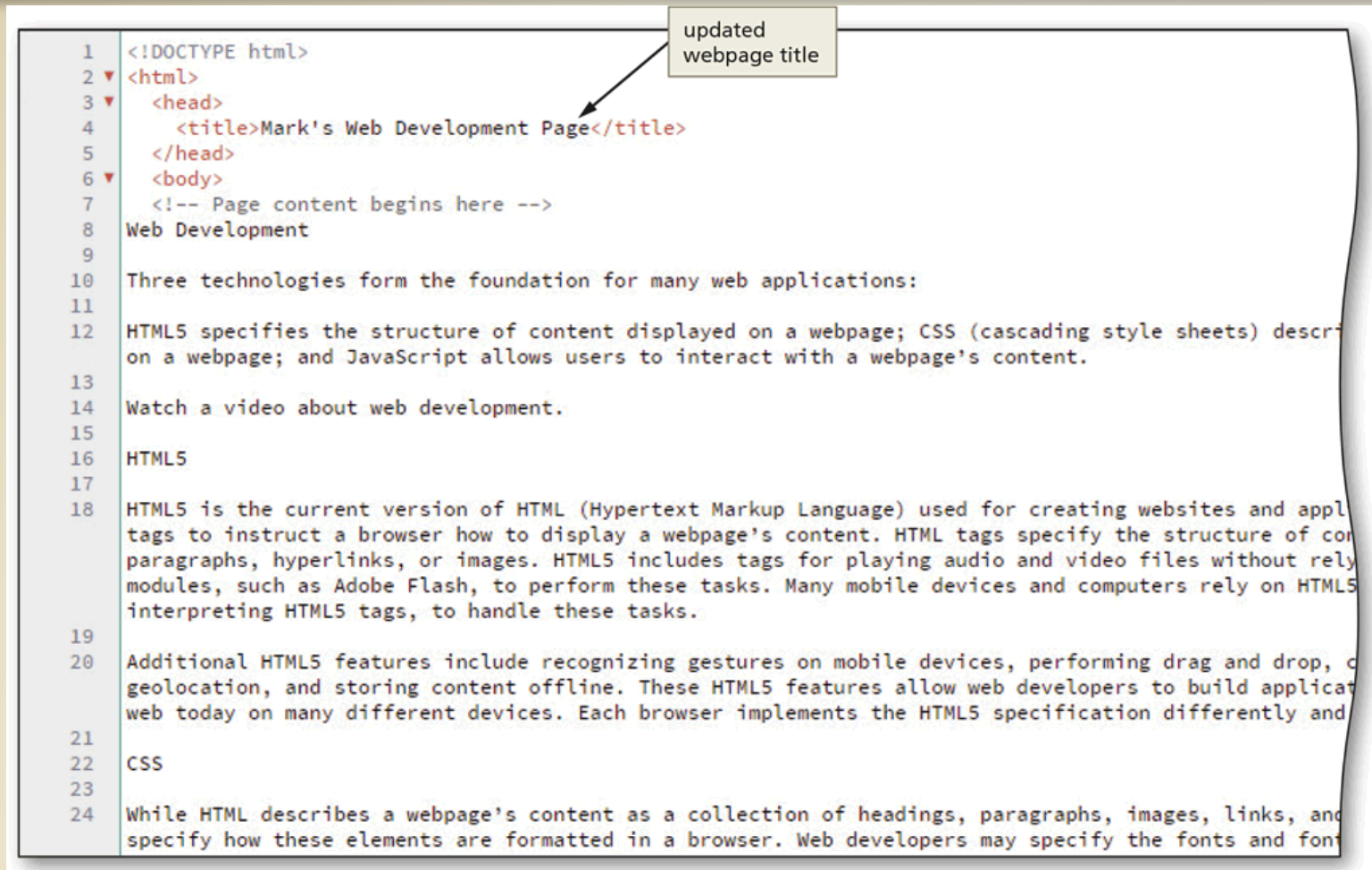
```
1 <!-- Page content begins here -->
2
3
4 <title>Page Title</title>
5 </head>
6 <body>
7 <!-- Page content begins here -->
8 Web Development
9
10 Three technologies form the foundation for many web applications:
11
12 HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describes the design and appearance of information
13 on a webpage; and JavaScript allows users to interact with a webpage's content.
14
15 Watch a video about web development.
16
17 HTML5
18 HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications. HTML uses a set of codes called
19 tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content on a webpage, such as headings,
20 paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying on the use of third-party plug-ins, or
21 modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5-compliant browsers, which are capable of
22 interpreting HTML5 tags, to handle these tasks.
23
24 Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop, creating graphics dynamically, accessing a
25 geolocation, and storing content offline. These HTML5 features allow web developers to build applications that meet the needs of people using the
26 web today on many different devices. Each browser implements the HTML5 specification differently and may not support all of its features.
27
28 CSS
29 While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and other elements, CSS allows web developers to
30 specify how these elements are formatted in a browser. Web developers may specify the fonts and font sizes, colors, borders, backgrounds, and
31 other styles to apply to each of these elements.
32
33 CSS provides web developers with precise control over a webpage's layout. With CSS, web developers can apply different layouts to the same
34 information so that it is formatted appropriately for printing or viewing on computers and devices with varying screen sizes. The current version
35 of CSS is known as CSS3 (cascading style sheets, version 3).
36
37 JavaScript
38 JavaScript is a programming language that adds interactivity to webpages. It often is used to check for appropriate values on web forms, present
39 alert messages, display menus on webpages, and create other dynamic content. Some developers use JavaScript to detect the user's browser version
40 in order to display a webpage especially designed for that browser.
41
42
43
```

starter text inserted

Adding the Webpage Title

- Select the text, Page Title, that appears between the `<title>` and `</title>` tags
- **Type** Mark's Web Development Page as the title. Replace the name, Mark, with your first name

Adding the Webpage Title



Headings

- Headings indicate the different sections of a webpage. HTML supports six levels of headings, which are identified by the following tags: `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`

Headings

```
<html>
  <head>
    <title>Headings</title>
  </head>
  <body>
    <h1>Heading level 1</h1>
    <h2>Heading level 2</h2>
    <h3>Heading level 3</h3>
    <h4>Heading level 4</h4>
    <h5>Heading level 5</h5>
    <h6>Heading level 6</h6>
  </body>
</html>
```

The diagram illustrates the mapping between HTML heading tags and their visual output. Arrows point from the closing tags in the code to the corresponding heading levels:

- `</h1>` points to **Heading level 1**
- `</h2>` points to **Heading level 2**
- `</h3>` points to **Heading level 3**
- `</h4>` points to **Heading level 4**
- `</h5>` points to **Heading level 5**
- `</h6>` points to **Heading level 6**

Identifying the Headings in the index.html File

- Position the insertion point at the beginning of the line that reads, Web Development
- Type `<h1>` to identify where the heading begins
- Position the insertion point at the end of the line of text that reads, Web Development
- Type `</h1>` to identify where the heading ends. Although the browser disregards spacing, you may type blank spaces to indent this tag so it lines up with its opening `<h1>` tag

Identifying the Headings in the index.html File

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Mark's Web Development Page</title>
5   </head>
6   <body>
7     <!-- Page content begins here -->
8     <h1>Web Development</h1>
9
10    Three technologies form the foundation for many web applications:
11
12    HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describes
13    on a webpage. JavaScript allows users to interact with a webpage's content.
14
15    Watch a video about web development.
16
17    HTML5
18
19    HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications.
20    HTML tags specify the structure of content displayed on a webpage. HTML tags specify the structure of content
21    paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying on
22    modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5
23    interpreting HTML5 tags, to handle these tasks.
24
25    Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop, geolocation,
26    and storing content offline. These HTML5 features allow web developers to build applications that work on
27    web today on many different devices. Each browser implements the HTML5 specification differently and
28
29    CSS
30
31    While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and
32    specific to these elements are formatted in a browser. Web developers may specify the fonts and colors
```

Identifying the Headings in the index.html File

- Position the insertion point at the beginning the line that reads, HTML5
- Type `<h2>` to identify where the heading begins
- Position the insertion point at the end of the line of text that reads, HTML5
- Type `</h2>` to identify where the heading ends. To increase readability, you may type spaces to indent this tag so it lines up with its opening `<h2>` tag

Identifying the Headings in the index.html File

```
2  ▼ <html>
3  ▼   <head>
4      <title>Mark's Web Development Page</title>
5   </head>
6  ▼   <body>
7      <!-- Page content begins here -->
8      <h1>Web Development</h1>
9
10     Three technologies form the foundation for many web applications:
11
12     HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describe
13     on a webpage; and JavaScript allows users to interact with a webpage's content.
14
15     Watch a video about web development.
16
17     <h2>HTML5</h2>
18     HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications.
19     tags tell a browser how to display a webpage's content. HTML tags specify the structure of content, such as
20     paragraphs, links, or images. HTML5 includes tags for playing audio and video files without relying on
21     modules like Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5
22     interpreting HTML5 tags, to handle these tasks.
23
24     Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop,
25     geolocation, and storing content offline. These HTML5 features allow web developers to build applications
26     web today on many different devices. Each browser implements the HTML5 specification differently and
27
28     CSS
29
30     While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and
31     specify how these elements are formatted in a browser. Web developers may specify the fonts and font
32     other styles to apply to each of these elements.
```

opening and closing <h2> tags

Identifying the Headings in the index.html File

- Repeat the previous steps to identify the lines that read, CSS and JavaScript, as `<h2>` headings, and indent each line by four spaces
- Save the changes to the index.html file

Identifying the Headings in the index.html File

```
5    </head>
6    <body>
7      <!-- Page content begins here -->
8      <h1>Web Development</h1>
9
10     Three technologies form the foundation for many web applications:
11
12     HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describe
13     on a webpage; and JavaScript allows users to interact with a webpage's content.
14
15     Watch a video about web development.
16
17     <h2>HTML5</h2>
18
19     HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications.
20     HTML tags instruct a browser how to display a webpage's content. HTML tags specify the structure of content,
21     paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying on
22     modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5
23     interpreting HTML5 tags, to handle these tasks.
24
25     Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop,
26     geolocation, and storing content offline. These HTML5 features allow web developers to build applications
27     web today on many different devices. Each browser implements the HTML5 specification differently and
28
29     <h2>CSS</h2>
30
31     While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and
32     specify how these elements are formatted in a browser. Web developers may specify the fonts and font
33     other styles to apply to each of these elements.
34
35     CSS provides web developers with precise control over a webpage's layout. With CSS, web developers can
36     information so that it is formatted appropriately for printing or viewing on computers and devices with
37     of CSS is known as CSS3 (cascading style sheets version 3).
38
39     <h2>JavaScript</h2>
40
41     JavaScript is a programming language that adds interactivity to webpages. It often is used to check
42     alert messages, display menus on webpages, and create other dynamic content. Some developers use JavaScript
43     in order to display a webpage especially designed for that browser.
```

<h2> tags surround CSS and JavaScript headings

Viewing the index.html Webpage in a Browser

- Minimize the text editor window showing the HTML code for the index.html file
- If necessary, navigate to the location of the index.html file
- Double-click the index.html file to open it in a browser
- When you are finished viewing the webpage in the browser, minimize the browser window and redisplay the text editor containing the index.html file

Viewing the index.html Webpage in a Browser

Web Development

heading created with <h1> tag

Three technologies form the foundation for many web applications: HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describes the design and appearance of information on a webpage; and JavaScript allows users to interact with a webpage's content. Watch a video about web development.

HTML5

HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications. HTML uses a set of codes called tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content on a webpage, such as headings, paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying on the use of third-party plug-ins, or modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5-compliant browsers, which are capable of interpreting HTML5 tags, to handle these tasks. Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop, creating graphics dynamically, accessing a geolocation, and storing content offline. These HTML5 features allow web developers to build applications that meet the needs of people using the web today on many different devices. Each browser implements the HTML5 specification differently and may not support all of its features.

headings created with <h2> tags

CSS

While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and other elements, CSS allows web developers to specify how these elements are formatted in a browser. Web developers may specify the fonts and font sizes, colors, borders, backgrounds, and other styles to apply to each of these elements. CSS provides web developers with precise control over a webpage's layout. With CSS, web developers can apply different layouts to the same information so that it is formatted appropriately for printing or viewing on computers and devices with varying screen sizes. The current version of CSS is known as CSS3 (cascading style sheets, version 3).

JavaScript

Paragraphs

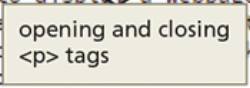
- The `<p>` and `</p>` tags are used to identify the beginning and ending of Paragraphs
- If you have several paragraphs of text on your webpage, these tags will inform the browser to insert additional line spacing above and below the paragraph so that the text is easier to read when displayed in the browser
 - The browser ignores line breaks and line spacing in the HTML file, so it is important to properly define the paragraphs using the `<p>` and `</p>` tags

Identifying the Paragraphs in the index.html File

- Position the insertion point at the beginning of the paragraph that begins with, Three technologies form the foundation....
- Type `<p>` to indicate where the paragraph begins
- Position the insertion point at the end of the first paragraph
- Type `</p>` to indicate the end of the paragraph. Although spacing does not matter to the browser, you may type spaces to indent this tag so it lines up with its opening `<p>` tag

Identifying the Paragraphs in the index.html File

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Mark's Web Development Page</title>
5   </head>
6   <body>
7     <!-- Page content begins here -->
8     <h1>Web Development</h1>
9
10    <p>Three technologies form the foundation for many web applications:</p>
11
12    HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style sheets) describes
13    on a webpage; and JavaScript allows users to interact with a webpage's content.
14
15    Watch a video about web development.
16
17    <h2>HTML5</h2>
18
19    HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications.
20    tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content,
21    paragraphs, hyperlinks, or images. There are also tags for playing audio and video files without relying on
22    modules, such as Adobe Flash, to handle these tasks. Many mobile devices and computers rely on HTML5
23    interpreting HTML5 tags, to handle these tasks.
24
25    Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop,
26    geolocation, and storing content offline. These HTML5 features allow web developers to build applications
27    web today on many different devices. Each browser implements the HTML5 specification differently and
28
29    <h2>CSS</h2>
30
31    While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and
32    specify how these elements are formatted in a browser. Web developers may specify the fonts and
33    other styles to apply to each of these elements.
```

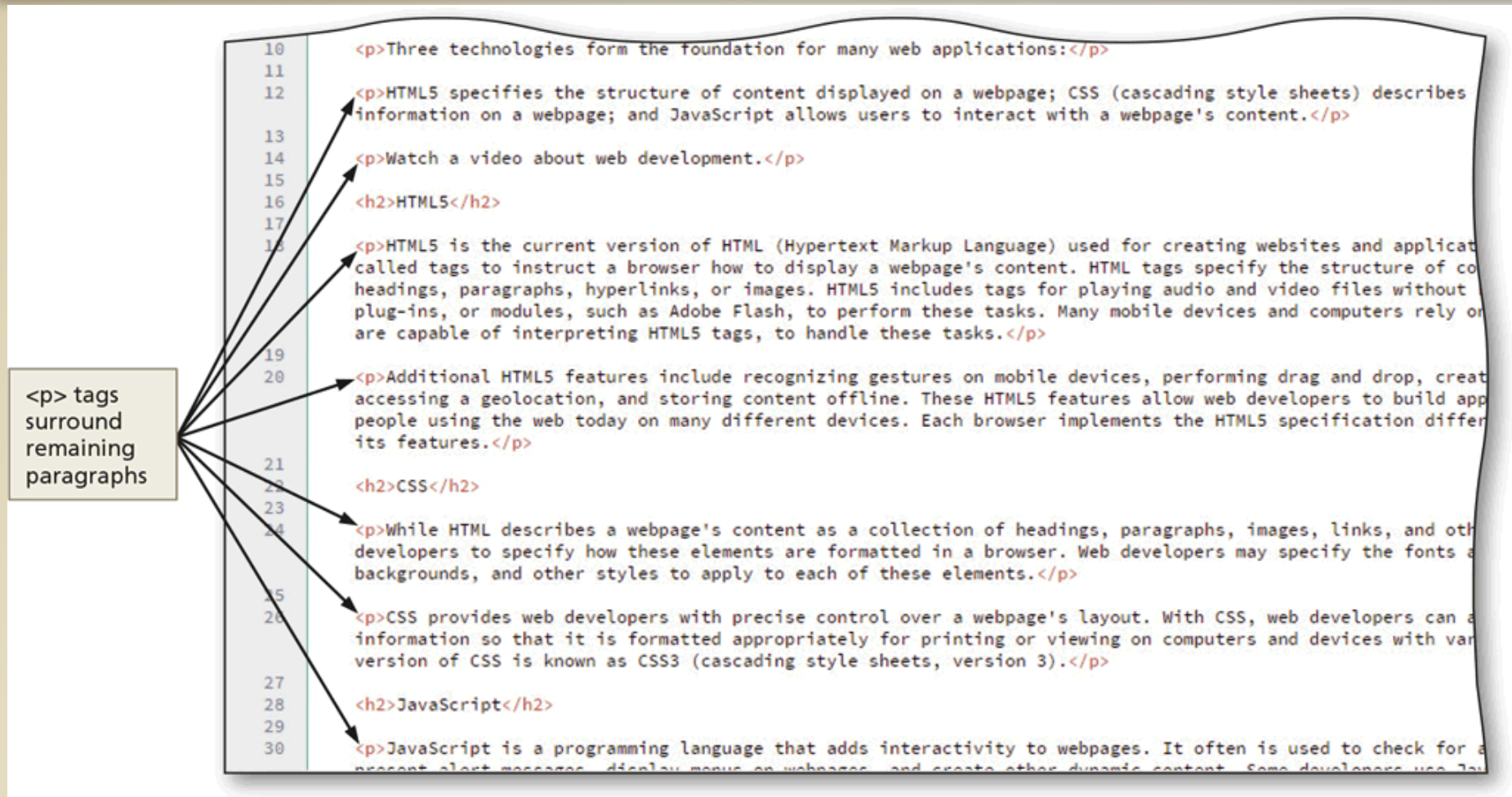


A diagram with a light beige background and a thin black border. It contains the text "opening and closing" on the top line and "<p> tags" on the bottom line. Two black arrows originate from this box: one points diagonally up and to the left towards the opening <p> tag on line 10 of the code, and the other points diagonally up and to the right towards the closing </p> tag on the same line.

Identifying the Paragraphs in the index.html File

- For the remaining paragraphs in the index.html file, type `<p>` at the beginning of each paragraph and type `</p>` at the end of each paragraph. To increase readability, you may type spaces to indent this tag so that it lines up with its opening `<p>` tag
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Identifying the Paragraphs in the index.html File



Images

- Images can be either photos or graphics
- Images always are stored as separate files, and references to the images appear in the HTML code using the `<img>` tag
- Common attributes for the `<img>` tag describe the location of an image file, alternate text for the image, and a style that indicates how to position the image

Inserting Images in the index.html File

- Locate the `<h2>HTML5</h2>` heading in the file. Position the insertion point before the first `<p>` tag that follows this heading and then press the ENTER key to insert a new line
- Type `` to insert a reference to the `html5_image.png` file in the images folder, set the alternate text to HTML5 Image, and style the image so that it is aligned to the left of the text under the HTML5 heading

Inserting Images in the index.html File

```
7  <!-- Page content begins here -->
8  <h1>Web Development</h1>
9
10 <p>Three technologies form the foundation for many web applications:</p>
11
12 <p>HTML5 specifies the structure of content displayed on a webpage; CSS (cascading style
13 information on a webpage; and JavaScript allows users to interact with a webpage's content
14
15 <p>Watch a video about web development.</p>
16
17 <h2>HTML5</h2>
18 
19
20 <p>HTML5 is the current version of HTML (Hypertext Markup Language) used for creating web
21 called tags to instruct a browser how to display a webpage's content. HTML tags specify the
headings, paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video
plug-ins, or modules, such as Adobe Flash, to perform these tasks. Many mobile devices are
are capable of interpreting HTML5 tags, to handle these tasks.</p>
20
21 <p>Additional HTML5 features include recognizing gestures on mobile devices, performing
accessing a geolocation, and storing content offline. These HTML5 features allow web developers
people using the web today on many different devices. Each browser implements the HTML5 spec
```

 tag
inserted

Inserting Images in the index.html File

- Locate the `<h2>CSS</h2>` heading in the file. Position the insertion point before the first `<p>` tag that follows this heading and then press the ENTER key to insert a new line
- **Type** `` to insert a reference to the `css_image.png` file in the images folder, set the alternate text to CSS Image, and style the image so that it is aligned to the right of the text under the CSS heading

Inserting Images in the index.html File

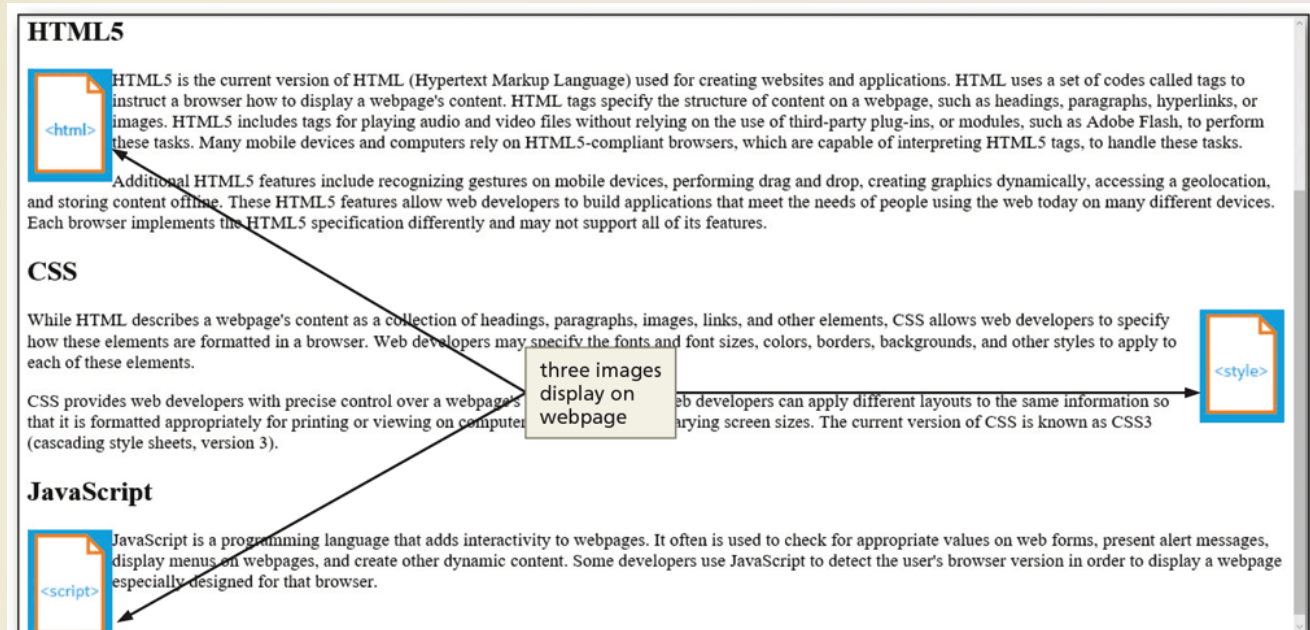
- Locate the `<h2>JavaScript</h2>` heading in the file. Position the insertion point before the first `<p>` tag that follows this heading and then press the ENTER key to insert a new line
- **Type** `` to insert a reference to the js_image.png file in the images folder, set the alternate text to JavaScript Image, and align the image to the left of the text under the JavaScript heading
- Save the changes to the index.html file

Inserting Images in the index.html File



Inserting Images in the index.html File

- Refresh or reload the webpage in the browser window to verify the changes are displayed properly. If necessary, scroll to display the three images



Links

- A link, or hyperlink, can be text or an image in a webpage that a user clicks to navigate to another webpage, download a file, or perform another action, such as running an email app and addressing an email message

HYPERLINK WITH ABSOLUTE REFERENCE

```
<a href="http://www.w3.org" target="_blank">HTML5</a>
```

HYPERLINK WITH RELATIVE REFERENCE

```
<a href="video.html"> Watch a video</a>
```

Inserting a Link with an Absolute Reference

- In the second paragraph beginning, HTML5 specifies the structure, position the insertion point immediately before the H in HTML5
- Type `<a href="http://www.w3.org/html/logo" target="_blank">` to specify the link destination and that the resulting webpage should open in a new, blank window
- Position the insertion point after HTML5 and then type `</a>` to indicate the end of the link

Inserting a Link with an Absolute Reference

```
8 <h1>Web Development</h1>
9
10 <p>Three technologies form the foundation for many web applications:</p>
11
12 <p><a href="http://www.w3.org/html/logo" target="_blank">HTML5</a> specifies the structure of content displayed o
(cascading style sheets) describes the design and appearance of information on a webpage; and JavaScript allows u
webpage's content.</p>
13
14 <p>Watch a video about web development.</p>
15
16 <h2>HTML5</h2>
17 
18
19 <p>HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications. HT
called tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content on
headings, paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying
plug-ins, or modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely on HTML5-
are capable of interpreting HTML5 tags, to handle these tasks.</p>
20
```

opening and closing <a> tags surround link text

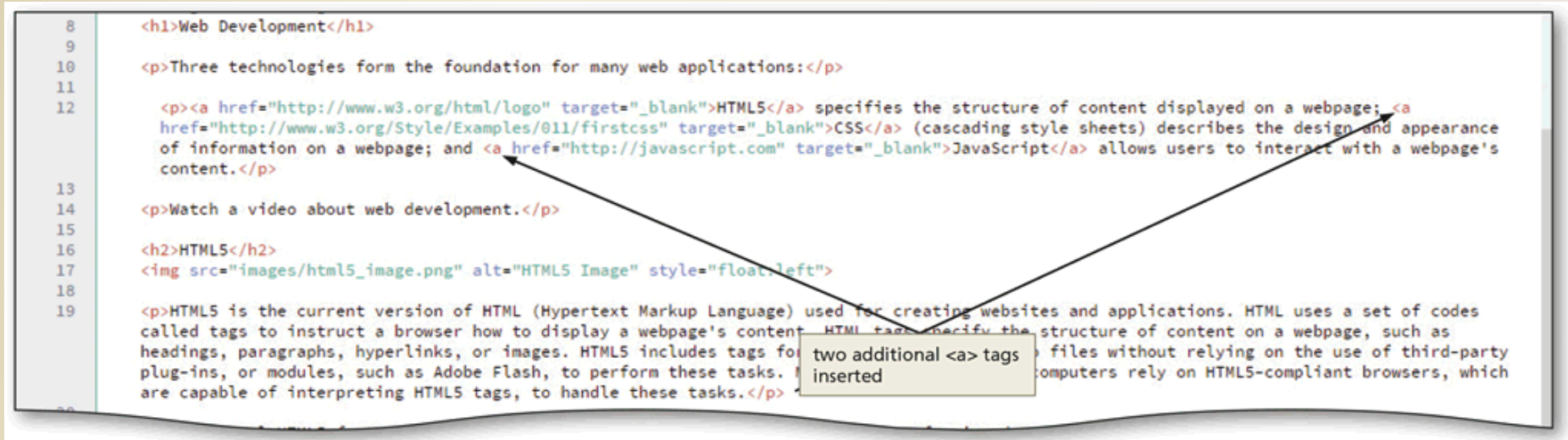
Inserting a Link with an Absolute Reference

- In the same paragraph, position the insertion point immediately before CSS and type `<a href="http://www.w3.org/Style/Examples/011/firstcss" target=" blank">` to specify the link destination and that the resulting webpage should open in a new, blank window
- Position the insertion point after CSS and then type `</a>` to indicate the end of the link
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Inserting a Link with an Absolute Reference

- In the same paragraph, position the insertion point immediately before JavaScript and type `<a href="http://javascript.com" target="_blank">` to specify the link destination and that the resulting webpage should open in a new, blank window
- Position the insertion point after JavaScript and then type `</a>` to indicate the end of the link
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Inserting a Link with an Absolute Reference



Creating the video.html File

- Open the template.html file in a text editor.
- Save the file with the file name, video.html.
- Between the `<title>` and `</title>` tags, select the existing text and then type `Web Development` to replace the text
- Save the changes to the video.html file. Do not close the file

Inserting a Link with a Relative Reference

- Display the index.html file
- Position the insertion point immediately before the W in the paragraph beginning, Watch a video
- Type `<a href="video.html">` to indicate the beginning of the link that will point to video.html, which is in the same folder as index.html
- Position the insertion point after the word, video, in the same sentence and then type `</a>` to indicate the end of the link
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Inserting a Link with a Relative Reference



Unordered and Ordered Lists

Unordered List

```
<ul>  
  <li>HTML5</li>  
  <li>CSS</li>  
  <li>JavaScript</li>  
</ul>
```

- HTML5
- CSS
- JavaScript

Ordered List

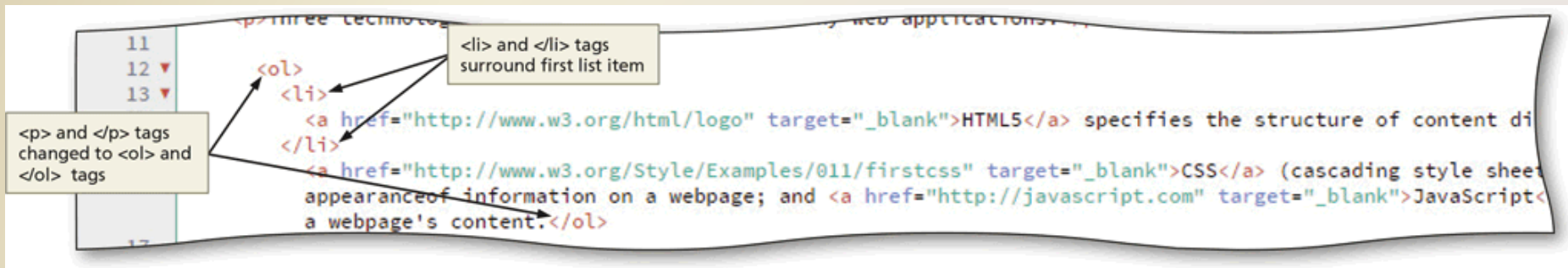
```
<ol>  
  <li>HTML5</li>  
  <li>CSS</li>  
  <li>JavaScript</li>  
</ol>
```

1. HTML5
2. CSS
3. JavaScript

Adding an Ordered List

- The items in this ordered list will include the three links for HTML5, CSS, and JavaScript. To change the first paragraph into an ordered list, change the first `<p>` to `<ol>` and the corresponding `</p>` to `</ol>`
- Position the insertion point before the `<a>` tag for the HTML5 link, press the ENTER key, press the SPACEBAR two times to create an indent, and then type `<li>` to indicate the beginning of a list item, and then press the ENTER key
- Position the insertion point immediately after the semicolon in the HTML5 sentence, press the ENTER key, press the SPACEBAR until the insertion point lines up with the opening `<li>` tag, and then type `</li>` to indicate the end of the first list item. Adjust the line spacing and indentation as necessary
- Press the ENTER key so that the next list item will begin on a new line in the file

Adding an Ordered List



Adding an Ordered List

- Insert the remaining list items, typing the `<li>` and `</li>` tags for the CSS and JavaScript sentences
- Adjust the line spacing and indentation
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Adding an Ordered List



The screenshot shows an HTML code editor with the following code:

```
8 <h1>Web Development</h1>
9
10 <p>Three technologies form the foundation for many web applications:</p>
11
12 <ol>
13 <li>
14 <a href="http://www.w3.org/html/logo" target="_blank">HTML5</a> specifies the structure of content displayed on a webpage;
15 </li>
16 <li>
17 <a href="http://www.w3.org/Style/Examples/011/firstcss" target="_blank">CSS</a> (cascading style sheets) describes the design and
18 appearance of information on a webpage; and
19 </li>
20 <li>
21 <a href="http://javascript.com" target="_blank">JavaScript</a> allows users to interact with a webpage's content.
22 </li>
23 </ol>
24
25 <p><a href="video.html">Watch a video</a> about web development.</p>
26
27 <h2>HTML5</h2>
28 
29
30 <p>HTML5 is the current version of HTML (Hypertext Markup Language) used for creating websites and applications. HTML uses a set of codes
31 called tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content on a webpage, such as
32 headings, paragraphs, hyperlinks, and so on. HTML5 also includes features for playing audio and video files, and for using third-part
```

A callout box on the left side of the editor, labeled "two additional list items", points to lines 19 and 20 of the code, which represent the two new list items being added to the ordered list.

Applying Styles with CSS

- Although it is possible to customize these elements of webpages using HTML, CSS makes it easier to specify the appearance of similar elements in the same webpage or same website
- Embedded styles, which define styles in the `<head>` section of the index.html document, apply to the entire webpage on which they are defined

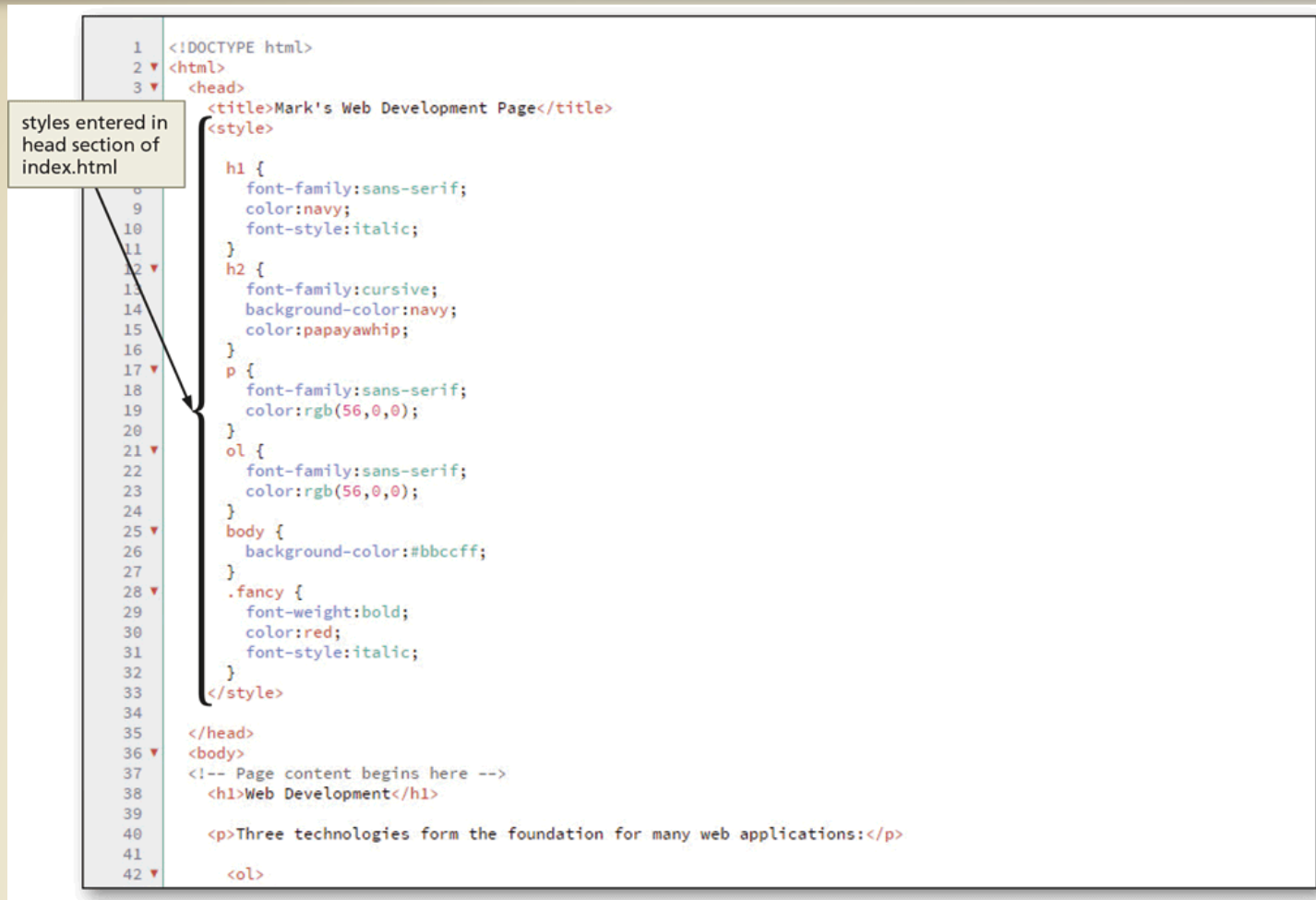
Applying Styles with CSS

```
<style>
  h1 {
    font-family:sans-serif;
    color:navy;
    font-style:italic;
  }
  h2 {
    font-family:cursive;
    background-color: navy;
    color:papayawhip;
  }
  p {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  ol {
    font-family:sans-serif;
    color:rgb(56,0,0);
  }
  body {
    background-color:#bbccff;
  }
  .fancy {
    font-weight:bold;
    color:red;
    font-style:italic;
  }
</style>
```

Adding CSS to the index.html File

- Position the insertion point at the end of the `</title>` tag and then press the ENTER key
- Type the text from the previous slide and in Figure 31. Make sure you pay attention to the spacing and indentation to maximize readability
- Save the changes to the webpage
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Adding CSS to the index.html File



styles entered in head section of index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Mark's Web Development Page</title>
5   <style>
6     h1 {
7       font-family:sans-serif;
8       color:navy;
9       font-style:italic;
10    }
11    h2 {
12      font-family:cursive;
13      background-color:navy;
14      color:papayawhip;
15    }
16    p {
17      font-family:sans-serif;
18      color:rgb(56,0,0);
19    }
20    ol {
21      font-family:sans-serif;
22      color:rgb(56,0,0);
23    }
24    body {
25      background-color:#bbccff;
26    }
27    .fancy {
28      font-weight:bold;
29      color:red;
30      font-style:italic;
31    }
32  </style>
33 </head>
34 <body>
35 <!-- Page content begins here -->
36 <h1>Web Development</h1>
37
38 <p>Three technologies form the foundation for many web applications:</p>
39
40 <ol>
```

Adding the fancy Style to the index.html File

- Position the insertion point immediately before the word, current, in the first paragraph below the HTML5 heading
- Type `<span class="fancy">` to indicate where you want to begin applying the fancy style
- Position the insertion point immediately after the word, HTML, in the same sentence
- Type `</span>` to indicate where you want to stop applying the fancy style

Adding the fancy Style to the index.html File



Adding the fancy Style to the index.html File

- Position the insertion point immediately before the word, Additional, in the second paragraph below the HTML5 heading
- Type `<span class="fancy">` to indicate where you want to begin applying the fancy style
- Position the insertion point immediately after the word, features, in the same sentence
- Type `</span>` to indicate where you want to stop applying the fancy style
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly

Adding the fancy Style to the index.html File

```
56 <h2>HTML5</h2>
57 
58
59 <p>HTML5 is the <span class="fancy">current version of HTML</span> (Hypertext Markup Language) used for creating websites and a
HTML uses a set of codes called tags to instruct a browser how to display a webpage's content. HTML tags specify the structure
on a webpage, such as headings, paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without
the use of third-party plug-ins, or modules, such as Adobe Flash, to perform these tasks. Many mobile devices and computers rely
compliant browsers, which are capable of interpreting HTML5 tags, to handle these tasks.</p>
60
61 <p><span class="fancy">Additional HTML5 features</span> include recognizing gestures on mobile devices, performing drag and drop
graphics dynamically, accessing a geolocation, and storing content offline. These HTML5 features allow web developers to build
that meet the needs of people using the web today on many different devices. Each browser implements the HTML5 specification differently
and may not support all of its features.</p>
62
63 <h2>CSS</h2>
64 
65
66 <p>While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and other elements, CSS allows
developers to specify how these elements are formatted in a browser. Web developers may specify the fonts and font sizes, color
backgrounds, and other styles to apply to each of these elements.</p>
67
68 <p>CSS provides web developers with the ability to control a webpage's layout. With CSS, web developers can create a different layout
```

 and tags surround additional text to be formatted with specified fancy style

JavaScript

- JavaScript is code that can be added to HTML documents to enhance the webpage by adding interactivity or dynamic content

```
<script>
function showDateAndTime() {
document.getElementById("current_date").innerHTML= Date();
}
</script>
```

Figure 35a Code to retrieve the current date and time.

```
<p id="current_date"></p>
```

Figure 35b Code to display the current date and time using the JavaScript code in the **<head>** section.

Using JavaScript to Add the Current Date and Time

- Position the insertion point after the `</style>` tag in the `<head>` section and then press the ENTER key two times
- Type the code shown in Figure 35a to enter the JavaScript code that retrieves the current date and time. Be sure to apply the same spacing, indentation, and use of uppercase and lowercase letters as shown in the figure
- Position the insertion point immediately after the `y` in the `<body>` tag. Press the SPACEBAR one time and then type `onload=showDateAndTime()` to specify that you want to run the JavaScript code when the webpage loads

Using JavaScript to Add the Current Date and Time



Using JavaScript to Add the Current Date and Time

- Position the insertion point immediately after the last `</p>` tag in the index.html file and then press the ENTER key two times
- Type the code shown in Figure 35b to specify where to display the current date
- Save the changes to the index.html file
- Refresh or reload the webpage in the browser window to verify the changes are displayed properly
- Refresh or reload the page again to verify that the JavaScript updates the date and time that is displayed

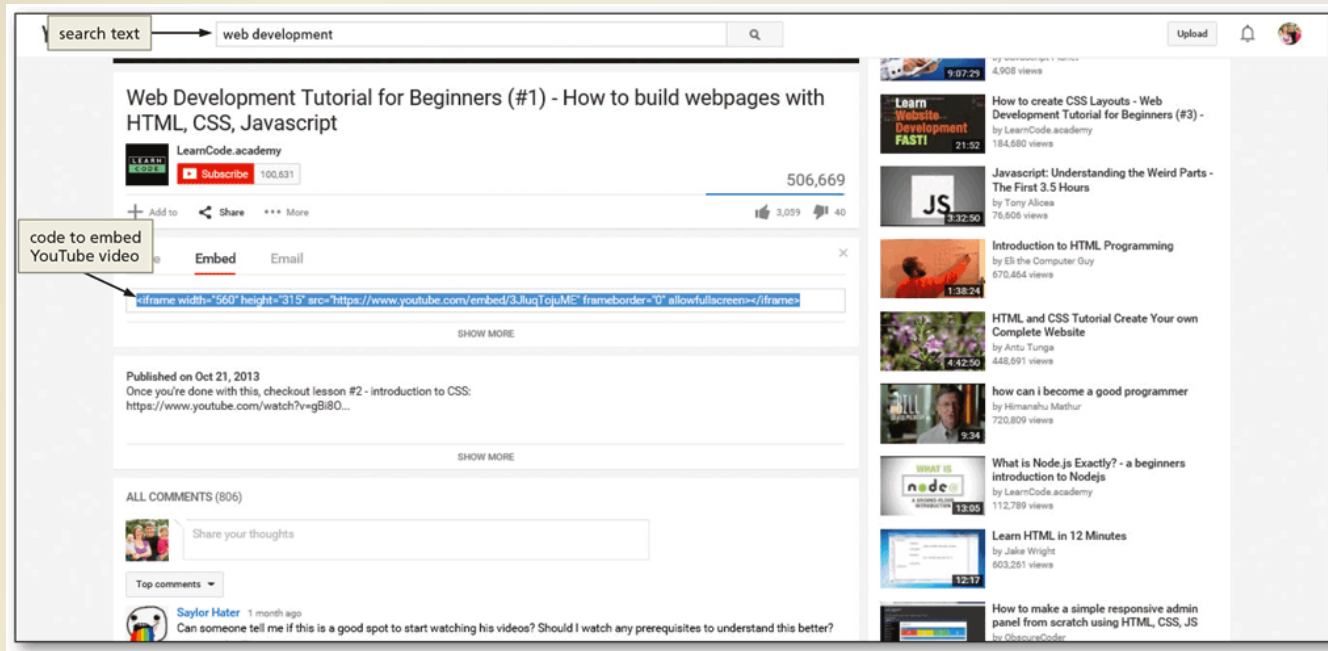
Using JavaScript to Add the Current Date and Time

```
67 <p><span class="fancy">Additional HTML5 features include recognizing gestures on mobile devices, performing background processing, creating
68 graphics dynamically, accessing a geolocation, and storing content offline. These HTML5 features allow web developers to build applications
69 that meet the needs of people using the web today on many different devices. Each browser implements the HTML5 specification differently and
70 may not support all of its features.</p>
71
72 <h2>CSS</h2>
73 
74
75 <p>While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and other elements, CSS allows web
76 developers to specify how these elements are formatted in a browser. Web developers may specify the fonts and font sizes, colors, borders,
77 backgrounds, and other styles to apply to each of these elements.</p>
78
79 <p>CSS provides web developers with precise control over a webpage's layout. With CSS, web developers can apply different layouts to the same
80 information so that it is formatted appropriately for printing or viewing on computers and devices with varying screen sizes. The current
81 version of CSS is known as CSS3 (cascading style sheets, version 3).</p>
82
83 <h2>JavaScript</h2>
84 
85
86 <p>JavaScript is a programming language that adds interactivity to webpages. It often is used to check for appropriate values on web forms,
87 present alert messages, display menus on webpages, and create other dynamic content. Some developers use JavaScript to detect the user's
88 browser version in order to display a webpage especially designed for that browser.</p>
89
90 <p id="current_date"></p>
91
92 <!-- Page content ends here -->
93 </body>
94 </html>
```

<p> tag
contains id
element to
display current
date and time

Adding a YouTube Video to a Webpage

- When you locate a video on that you want to include on a webpage, YouTube provides HTML code you can use to add the video to the webpage



Embedding a YouTube Video in the video.html Webpage

- Open a new browser tab and navigate to youtube.com
- Use the search box on the youtube.com webpage to locate a video about web development
- Locate the embed code. *Hint:* click the Share link and then click the Embed link
- Select the embed code in its entirety and then copy it to the clipboard

Embedding a YouTube Video in the video.html Webpage

- Display the text editor containing the code for the video.html webpage
- Position the insertion point on a blank line immediately following the line that reads,
`<!--Page content begins here -->`
- Paste the contents of the clipboard

Embedding a YouTube Video in the video.html Webpage

code to embed
YouTube video
copied from
YouTube website

```
html>  
<title>Web Development</title>  
5 </head>  
6 <body>  
7 <!-- Page content begins here -->  
8 <iframe width="560" height="315" src="https://www.youtube.com/embed/3JluqTojuME" frameborder="0" allowfullscreen></iframe>  
9  
10  
11 <!-- Page content ends here -->  
12 </body>  
13 </html>
```

Adding a Link Back to the index.html Webpage

- Position the insertion point on the blank line after the code pasted in from YouTube, immediately above the `</html>` tag, and then press the ENTER key
- Type `<br>` to add a line break after the code to display the video
- Press the ENTER key two times and then type `<a href="index.html">Home</a>` to add a link back to the website's home page referenced in index.html

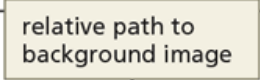
Adding a Link Back to the index.html Webpage



Adding a Background Style to the video.html Webpage

- When choosing a background image for a webpage, be sure to choose one that does not detract from the webpage content
- Background images create a pattern behind a webpage's content

```
<style>
  body {
    background-image:url("images/stripe_background.png");
  }
</style>
```



relative path to background image

The diagram shows a callout box with the text "relative path to background image" and an arrow pointing to the URL "images/stripe_background.png" in the CSS code. A bracket is also drawn under the entire URL string.

Adding a Background Style to the video.html Webpage

- Position the insertion point immediately after the `</title>` tag and then press the ENTER key
- Type the code in Figure 41. Be sure to apply the same line spacing and indenting as shown in the figure
- Save the changes to the video.html file

Adding a Background Style to the video.html Webpage



Exiting the Text Editor and Previewing the Webpages

- Close the video.html file in the text editor. If necessary, save the changes to the file
- Close the index.html file in the text editor. If necessary, save the changes to the file
- Redisplay the index.html file in the browser window. Click the Reload or Refresh button to make sure you are viewing the most current version of the webpage

Exiting the Text Editor and Previewing the Webpages

Web Development

'Watch a video' link

the foundation for many web applications:

1. [HTML5](#) specifies the structure of content displayed on a webpage;
2. [CSS](#) (cascading style sheets) describes the design and appearance of information on a webpage; and
3. [JavaScript](#) allows users to interact with a webpage's content.

[Watch a video](#) about web development.

HTML5



HTML5 is the **current version of HTML** (Hypertext Markup Language) used for creating websites and applications. HTML uses a set of codes called tags to instruct a browser how to display a webpage's content. HTML tags specify the structure of content of headings, paragraphs, hyperlinks, or images. HTML5 includes tags for playing audio and video files without relying on the use of third-party plug-ins, or modules, such as Adobe Flash, to do so. Mobile devices and computers rely on HTML5-compliant browsers, which are capable of interpreting HTML5 tags, to handle these tasks.

Additional HTML5 features include recognizing gestures on mobile devices, performing drag and drop, creating graphics dynamically, accessing a geolocation, and storing content offline. These HTML5 features allow web developers to build applications that meet the needs of people using the web today on many different devices. Each browser implements the HTML5 specification differently and may not support all of its features.

text formatted
in fancy style

CSS

While HTML describes a webpage's content as a collection of headings, paragraphs, images, links, and other elements, CSS allows web developers to specify how these elements are formatted in a browser. Web developers may specify the fonts and font sizes, colors, borders, backgrounds, and other styles to apply to each of these elements.

CSS provides web developers with precise control over a webpage's layout. With CSS, web developers can apply different layouts to the same information so that it is formatted appropriately for printing or viewing on computers and devices with varying screen sizes. The current version of CSS is known as CSS3 (cascading style sheets, version 3).



JavaScript



JavaScript is a programming language that adds interactivity to webpages. It often is used to check for appropriate values on web forms, present alert messages, display menus on webpages, and create other dynamic content. Some developers use JavaScript to detect the user's browser version in order to display a webpage especially designed for that browser.

Sat Aug 12 2017 13:08:23 GMT-0400 (Eastern Daylight Time)

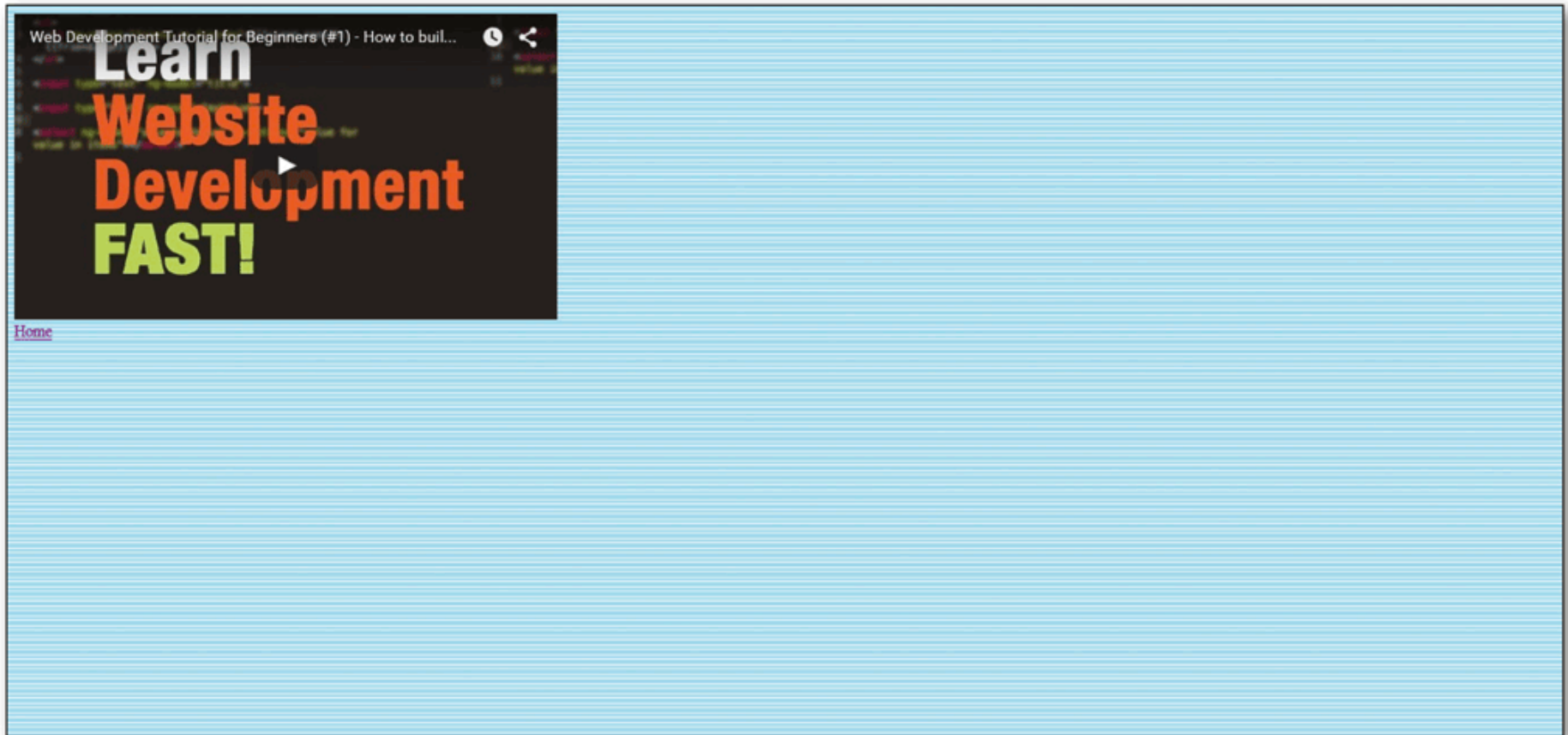
Exiting the Text Editor and Previewing the Webpages

- Verify the webpage you are viewing looks the same as Figure 43. The headings and paragraphs should be formatted, the page should have a light blue background color, two phrases should be formatted with the fancy style, and the three images should have loaded
- Click the HTML5 link to make sure it navigates to the proper destination. Then, return to the browser tab or window displaying the index.html webpage

Exiting the Text Editor and Previewing the Webpages

- Click the CSS link to make sure it navigates to the proper destination. Then, return to the browser tab or window displaying the index.html webpage
- Click the JavaScript link to make sure it navigates to the proper destination. Then, return to the browser tab or window displaying the index.html webpage
- Click the 'Watch a video' link to display the video.html webpage

Exiting the Text Editor and Previewing the Webpages



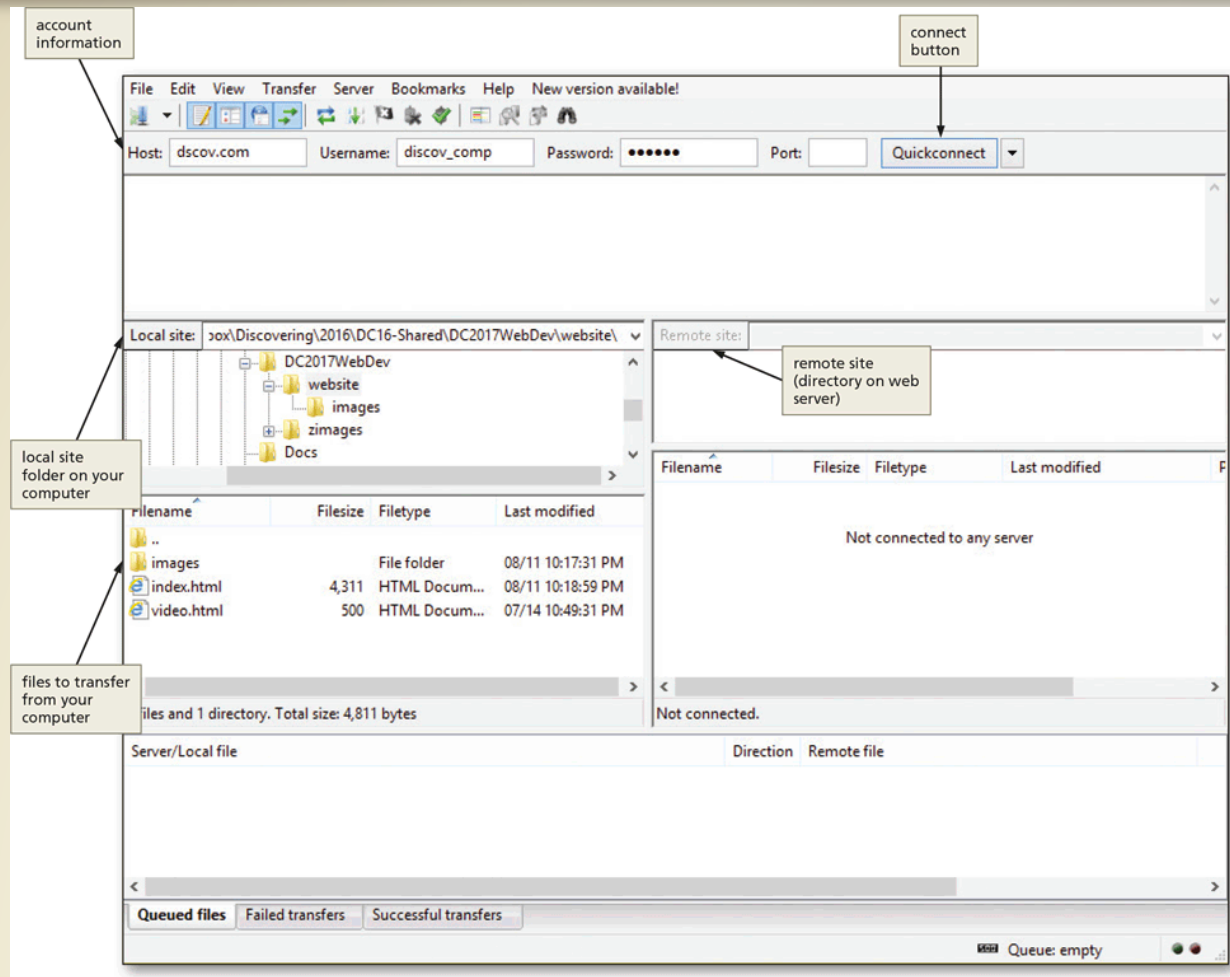
Exiting the Text Editor and Previewing the Webpages

- Click the play button in the video player to play the embedded video
- When you have finished watching the video, click the Home link at the bottom of the webpage to return to the index.html webpage
- Close all browser windows

Publishing a Website Online

- Type the host name (the name of the web server) and the user name and password for the account, or set up a profile containing this information using an FTP application
- Click the connect button to connect to the server

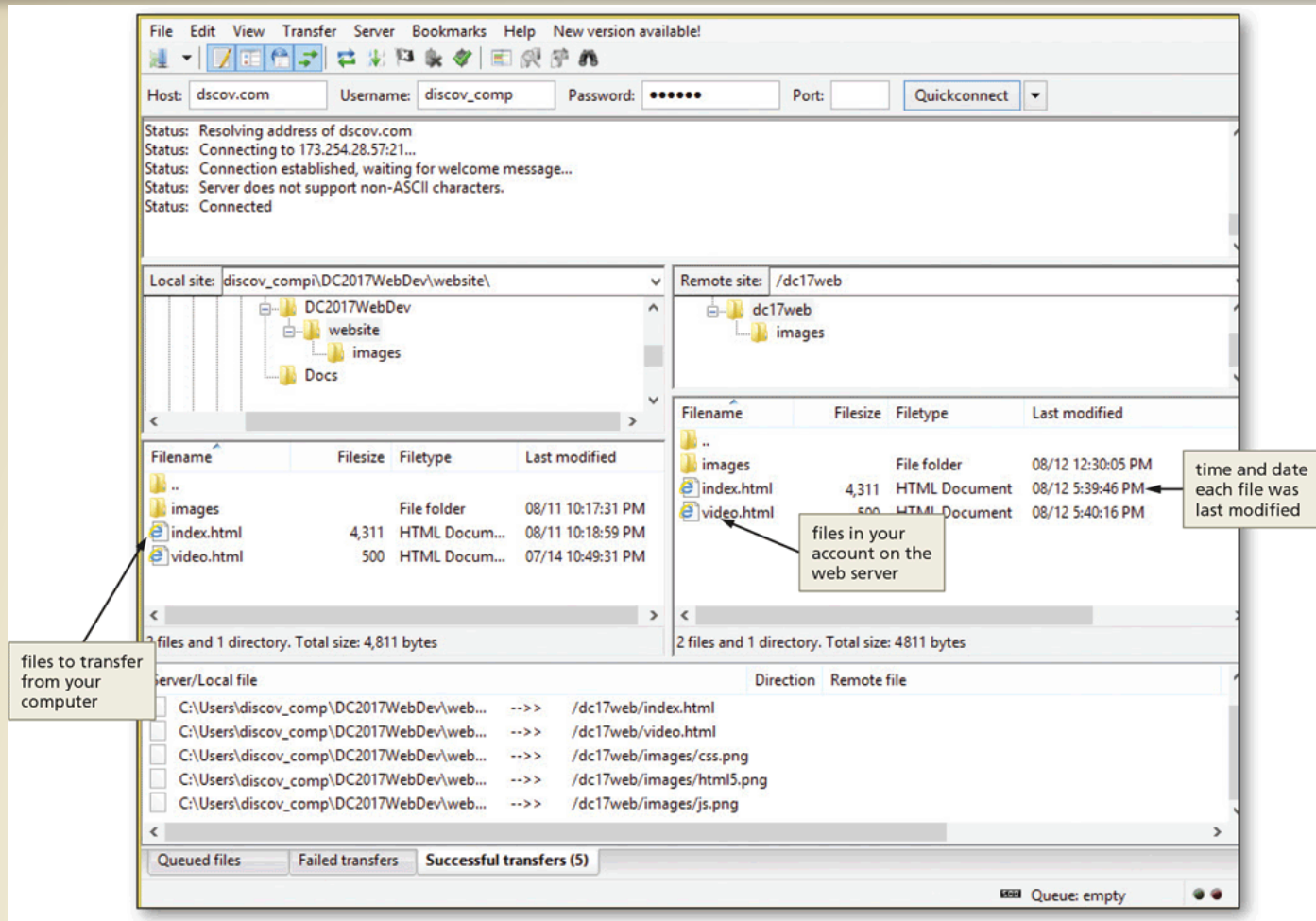
Publishing a Website Online



Publishing a Website Online

- In the local site section of the FTP application, navigate to the website folder containing the HTML, images, and other files for the website.
- In the remote site section of the FTP application, you should see the contents of your account on the web server. No files should appear if you have not yet uploaded any
- Select the index.html and video.html files and the images folder from the local websites section
- Drag the selected items to the remote website section, or select the upload option to upload these files to the web server

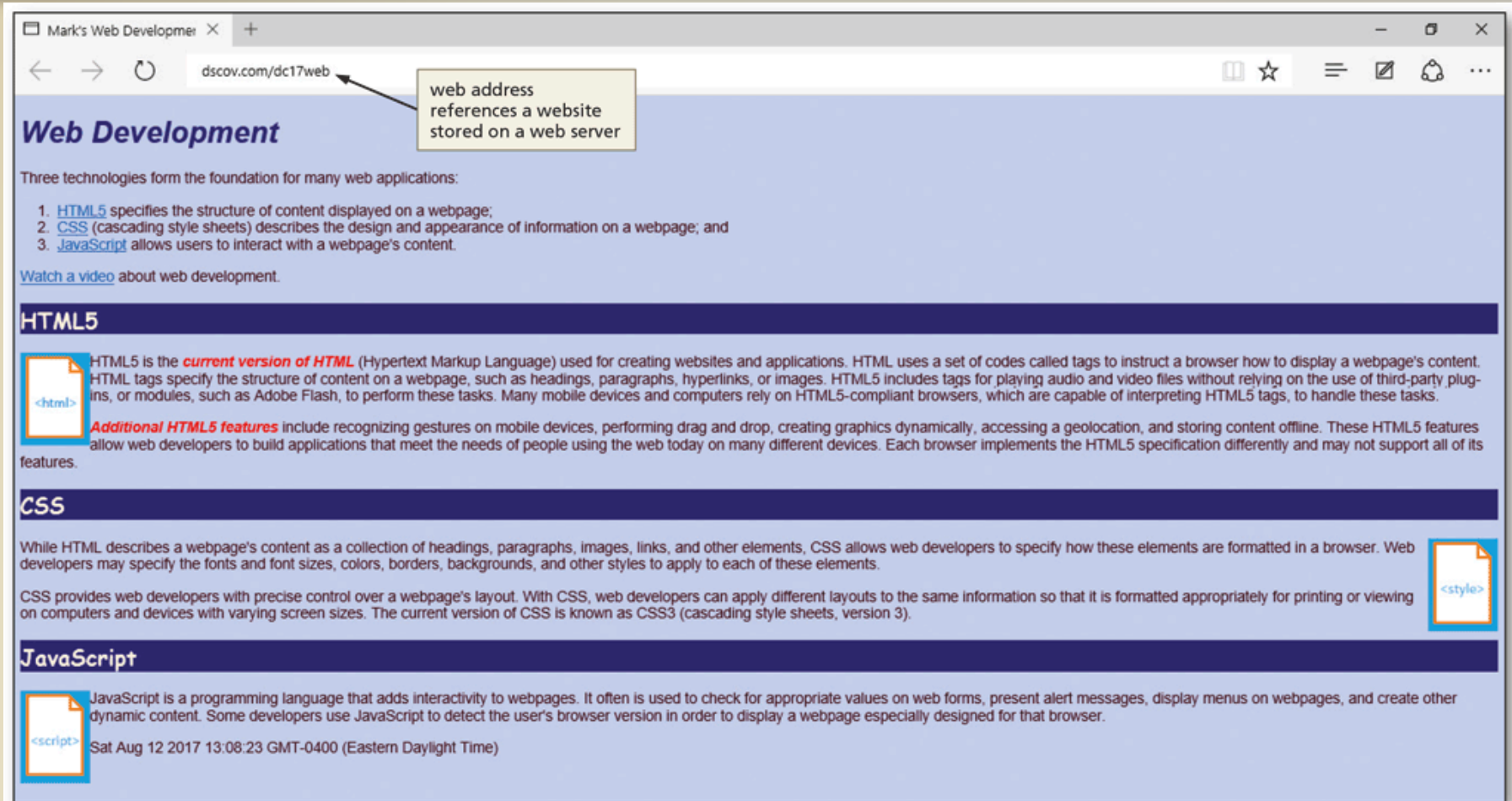
Publishing a Website Online



Viewing a Website Online

- Open a new browser or browser tab.
- In the address bar, type the absolute address of the website hosted on a web server. (This begins with `http://` and includes the name of the server hosting the site.) The website will appear

Viewing a Website Online



Summary of Tags and Styles

Table 2 Selected HTML5 Tags		
Tag	Example	Description
<code><!--></code>	<code><!--This is a comment. --></code>	Comment from web developer, ignored when page is rendered
<code><a></code>	<code>Google</code>	Anchor tag, specifies a link; specify href (hypertext reference) and use the attribute target="_blank" to display the page in a new tab
<code><body></code>	<code><body style="background-color:yellow"></code>	Body section of a webpage, styled to have a yellow background
<code>
</code>	<code>
</code>	Line break; <code>
</code> has no closing tag
<code><h1></code> through <code><h6></code>	<code><h1> This is a heading.</h1></code>	Headings for content; <code><h1></code> is largest, <code><h6></code> is smallest
<code><head></code>	<code><head> ... </head></code>	Head section
<code><hr></code>	<code><hr style="background-color:rgb(192,192,192);"></code>	Displays a horizontal rule (line) across the page to separate sections of content; optional style attributes may specify the background color or width of the line; <code><hr></code> has no closing tag
<code><html></code>	<code><html> ... </html></code>	Starts an HTML document
<code><iframe></code>	<code><iframe src="http://cengage.com" width="600" height="400"></iframe></code>	Includes content from another website, such as embedding webpage or an online video; use height and width attributes to specify the size, in pixels, of the iframe

Summary of Tags and Styles

<code></code>	<code></code>	Image tag, src attribute specifies the source or location of the image, alt attribute (required in HTML5) provides an alternate description of the image, height and width specify the display size of the image in pixels; <code></code> has no closing tag
<code></code>	<code> Item 1 Item 2 </code>	List item, used within <code></code> or <code></code> tags
<code></code>	<code> Item 1 Item 2 </code>	Ordered (numbered) list
<code><p></code>	<code><p>This is a paragraph.</p></code>	Paragraph
<code><script></code>	<code><script> ... </script></code>	Identifies JavaScript code; located in <code><head></code> section
<code></code>	<code>This text is formatted fancy.</code>	Identifies content to apply a custom style
<code><style></code>	<code><style> h1 { font-family:serif; color: blue; } </style></code>	Identifies embedded styles for tags; located in <code><head></code> section
<code><title></code>	<code><title>My Website</title></code>	Title of a webpage that appears in the browser tab; located in <code><head></code> section of document
<code></code>	<code> Item 1 Item 2 </code>	Unordered (bulleted) list

Summary of Tags and Styles

Table 3 Selected Styles		
Style	Example	Description
background-color	background-color:yellow;	Specifies the background color of elements, such as <code><p></code> , <code><h1></code> , and <code><body></code>
background-image	background-image:url("images/stripes.jpg")	Sets the background image of a <code><body></code> , <code><p></code> , <code><h1></code> , and other elements to the file whose path is given in the <code>url()</code> function
border	border: 3 px red;	Specifies a 4-sided border that is 3 pixels thick
color	color:blue;	Colors can be a web color name, a hexadecimal value, such as <code>#0000FF</code> , or an rgb value, such as <code>rgb(0,0,255)</code> that specifies the red, green, and blue components of the color
float	float:left;	Specifies whether to place an element to the left or right relative to text; often used to position an image to the left or right of text.
font-family	font-family:serif;	Specifies the font for a paragraph, heading, or other text element; use specific font names, such as <code>times</code> , <code>arial</code> , and <code>courier</code> , or family names, such as <code>serif</code> , <code>sans-serif</code> , <code>cursive</code> , or <code>monospace</code>
font-size	font-size:10px;	Specifies font size in pixels
font-style	font-style:italic;	Specifies font style; use normal, italic, or oblique as the value for this style
font-weight	font-weight:bold;	Specifies the font weight for a paragraph, heading, or other text element; use <code>bold</code> for thick characters, or numeric values 100 through 900, in increments of 100; 400 is the same as <code>normal</code> , 700 is the same as <code>bold</code>
text-align	text-align:left;	Sets alignment for <code><h1></code> , <code><h2></code> , or <code><h3></code> tags; values can be <code>left</code> , <code>center</code> , or <code>right</code>

Summary

Discuss tools for creating a website, such as text editors, code editors, and content management systems, and when to use each

Explain the uses of HTML5, CSS, and JavaScript technologies when creating websites

Discuss concepts related to web development, including static and dynamic content, relative and absolute references, HTML tags and attributes, and embedded and inline styles

Explain how to view a webpage's source code after displaying the page in a browser

Summary

Explain the unique role of the index.html page in a website

Use HTML tags to add a title, headings, paragraphs, images, links, ordered and unordered lists, and videos to a webpage

Use CSS to specify fonts, colors, and styles for text and background images or colors for webpages

Use JavaScript to display the current date and time on a webpage

Upload a website to a web server using an FTP program

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Focus On Web Development

Complete

